

Session #570

Developing Enterprise Applications for Palm Computing® Platform Devices

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version 1.2

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Developing Enterprise Applications



- **Section 1: Architecture Overview**
- **Section 2: Development Walkthrough**

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Architecture Overview

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Overview



- Extend enterprise (existing or new) data and services to mobile users
- Be able to update data using different networks
- Must support multiple PDA platforms

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Major Components

- **Enterprise application and database servers**
- **The network**
- **Mobile devices**



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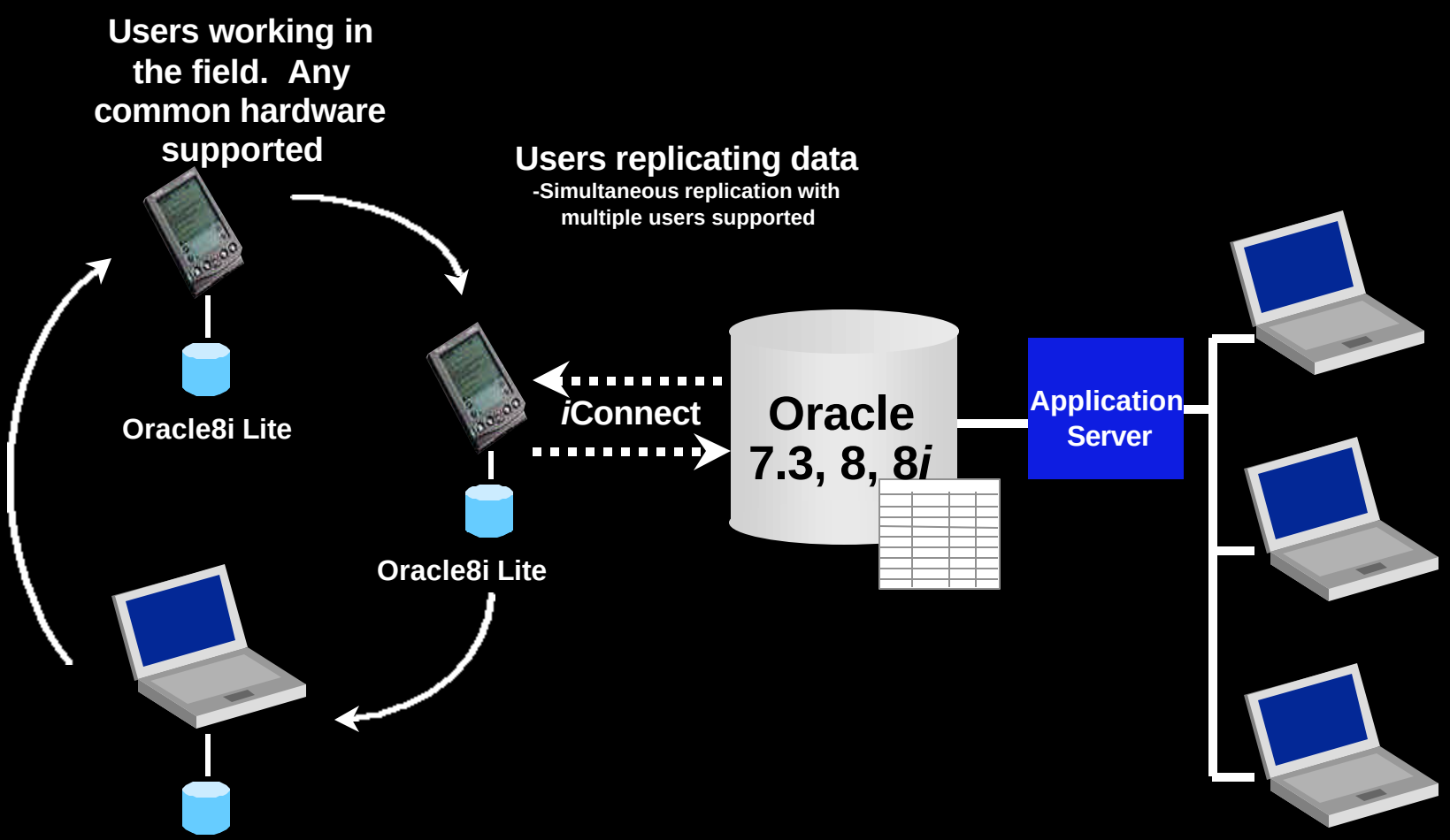
About Oracle8*i* Lite



- **An infrastructure that allows mobile devices to access their enterprise services and data**
- **Easily move existing data and schema to mobile platforms**
- **Supports all the popular PDA devices**
- **Replicates data and applications using different wired or wireless networks**

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Mobile Applications



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Oracle8*i* Lite (Cont.)



- **Includes:**
  - **A small footprint database engine on the device**
  - **Replication client (Consolidator Client)**
  - **Replication server (Consolidator Server)**
  - **Oracle 8i as the backend database**
- **Synchronize data with existing Oracle 8i databases**

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On the Client Device



- **Oracle Lite Database**
  - **Object-oriented database engine**
  - **Same interface across all platforms**
  - **Supports maintained index, complex query, transaction, and other database functionality**
  - **Very little execution overhead**
  - **Small footprint (~50Kbytes)**

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On the Client Device (Cont.)



- **Replication Client (Consolidator Client)**
  - **Replicates Oracle Lite database**
  - **Install and upgrade software (application deployment)**
  - **Can replicate native data files (.pdb)**
  - **Supports multiple transports**
    - ***HotSync***
    - ***HTTP/TCP/IP***
    - ***ScoutSync***

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On the Server



- **Consolidator Server**
  - **Publish and Subscribe Model**
  - **Synchronous bi-directional replication using subquery snapshots**
  - **Works with different transports**

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Supported Transport



- **HTTP/TCP/IP (using either Oracle Application Server or Web-To-Go Server)**
- **HotSync**
- **ScoutSync**
- **Others**

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Portability



- **Supported Devices**
  - **Palm Computing® Platform**
  - **EPOC**
  - **Windows CE**
  - **Windows (on laptops and desktops)**
- **Same database interface for all platforms**
- **Platform independent data**

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Scalability



- **Supports both 2-tier and 3-tier configurations**
- **Can support many mobile devices with one server**
- **Little extra overhead for each new account**

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Development Walkthrough

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Setup the Server



- **Setup transport(s)**
  - **HotSync®**
  - **HTTP**
  - **ScoutSync**
- **Create Publication Items**
  - **Define snapshots from existing (or new) tables**
- **Create User**
  - **Create user account**
  - **Select publication items for users**

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Client Side



- **Setup transport on device**
  - **TCP/IP setup**
- **Install Consolidator Client**
  - **HotSync consolidator client to device**
- **Run Consolidator Client to sync in sample data**

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Writing the Mobile Applications



- **Select development tools**
- **Study the sample projects**
  - **okDemo**
  - **okTest**
- **Write user interface code**
- **Access the database**

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Development Tools

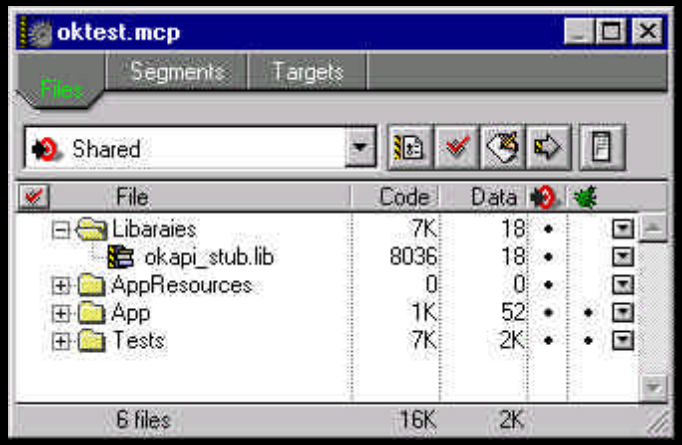


- **Metrowerks CodeWarrior**
- **GNU gcc**
- **PUMA Satellite Forms**
- **Others**

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Creating the Project

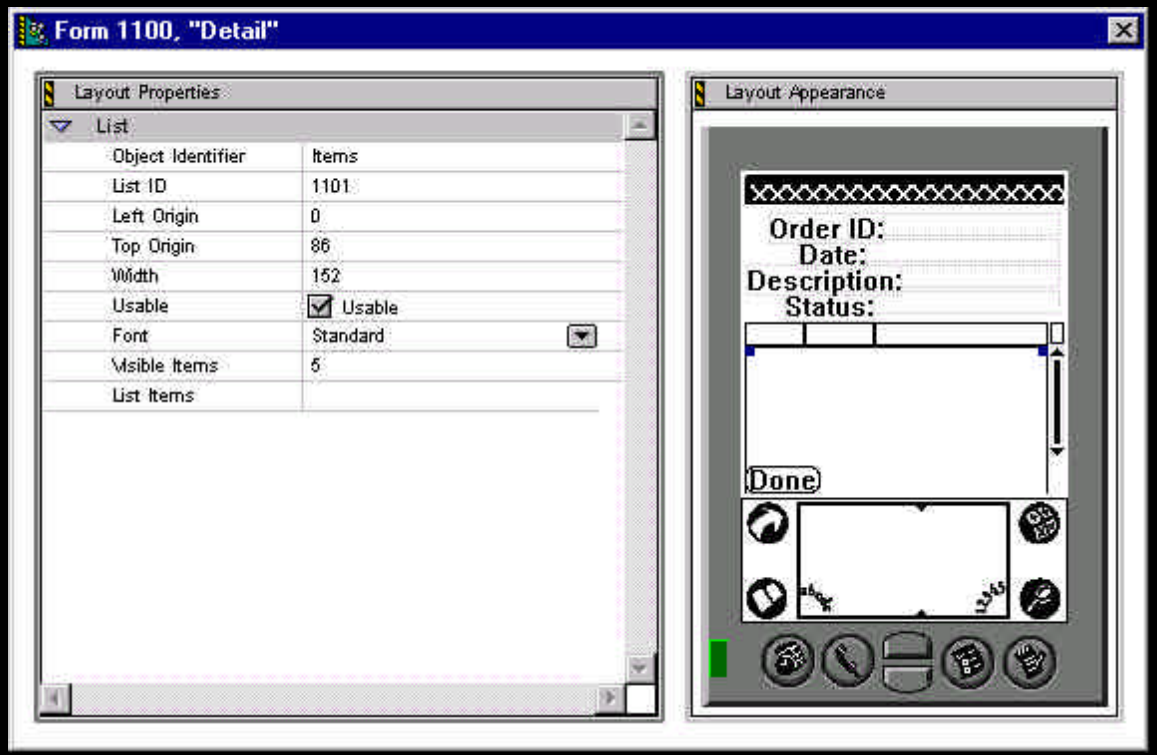
- Use existing project as template
- Include Oracle Lite header files in search path
  - (ORACLE_HOME/lite/pda40/palm/sdk/include)
- Add Oracle Lite libraries to project
  - Either shared library stubs or static libraries



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The User Interface

- **Define the list for forms and their usage**
- **List data required by each form**



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Accessing the Data



- **Create environment**
  - **okInit**
- **Connect to database**
  - **okConnect**
- **Get Schema Information**
  - **okFindClass and okFindGroup**

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Accessing the Data (Cont.)



- **Query**
  - **Build search conditions**
  - **Create iterator (okCreateIterator)**
  - **Iterator through result set (okIterate)**
  - **Get result object data (okGetAttrVal(s), okGetAttrValAt, okxLockObject + okxGetAttr + okxUnlockObject)**
  - **Built-in index optimizer**

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Accessing the Data (Cont.)



- **Updating objects**
  - **Set attributes (okSetAttrVal, okSetAttrValAt)**
  - **Indexes are maintained automatically**
- **Inserting new data**
  - **Creating objects (okCreateObj(s))**
- **Committing**
  - **Commit or rollback (okTransact)**

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Testing



- **Test application functions**
- **Test data updates from and to server**

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Deployment



- **Use Consolidator to deploy applications to mobile devices**
- **Use Oracle8i Lite Navigator utility to create snapshots and manage application deployment**
- **Upgrade new version**

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The Future



- **On-line / Off-line Modes**
  - **Transparent to application developers and users**
  - **Fetch required data as they are needed (information on demand)**
  - **Supports both connection based and connection broadcast updates**
- **Query Language**
  - **Support for standard query languages on PDAs**
- **Java Interface**
  - **Database Interface for Java programs**
  - **Java Stored Procedures support**

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Summary



- **Oracle8*i* Lite provides a well-defined and reliable architecture for building enterprise class mobile applications**
- **Reduces development, debugging and testing time**
- **Provides reliable and scalable access to enterprise data to mobile users**

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Questions



- **Any questions?**

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Comments



- **Please send us your comments**
  - **infolite@us.oracle.com**
  - **Newsgroup on OTN, better still**
- **Further information and sample code, demos available on OTN (Oracle Technical Network)**
  - **http://technet.oracle.com**

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Thanks :)

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